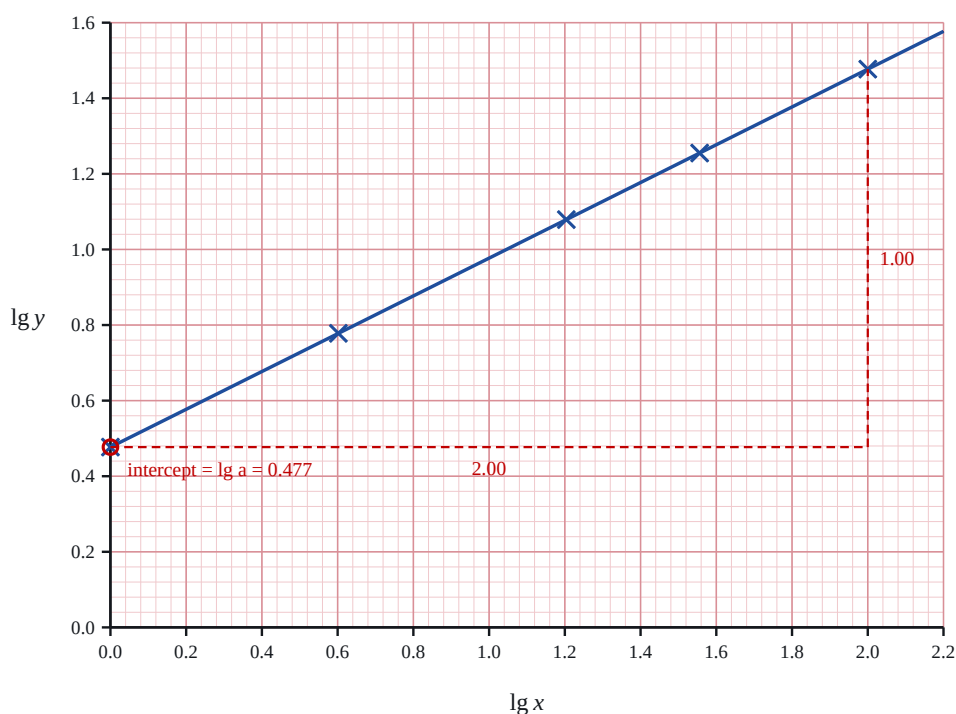


- 9 The variables x and y are related by $y = ax^b$, where a and b are constants. (a) Explain how a straight line graph may be drawn to represent this equation, and draw it for the given data on the grid below. [3]



Gradient $b = 1.00 \div 2.00 = 0.5$ · vertical intercept $\lg a = 0.477$, so $a = 10^{0.477} = 3$ · the ruled line is $\lg y = 0.477 + 0.5 \lg x$

WORKING — BLUE INK

$$y = ax^b \Rightarrow \lg y = \lg a + b \lg x$$

Comparing with $Y = mX + c$: plot $\lg y$ against $\lg x$;
gradient b , vertical intercept $\lg a$.

$\lg x$	0	0.602	1.204	1.556	2.000
$\lg y$	0.477	0.778	1.079	1.255	1.477

MARKS & NOTES — RED INK

Key concept: take logarithms of *both* variables — a power law needs $\lg x$, not x .

- **B1** $\lg y = \lg a + b \lg x$
- **B1** states: plot $\lg y$ against $\lg x$, gradient b , intercept $\lg a$
- **B1** both logarithm rows computed to 3 d.p., all five points plotted and a ruled straight line drawn
- **Trap** plotting $\lg y$ against x suits $y = ab^x$, not $y = ax^b$
- **Note** the intercept is $\lg a$, not a — part (b) still needs $a = 10^{0.477}$
- **Note** the points are exactly collinear (the data comes from $y = 3\sqrt{x}$), so any visible scatter means a logarithm was mis-read